MealVitals -- Market-to-Table Dietary Management

One in three Singapore residents struggle with chronic hyperlipidaemia (excessive cholesterol) or hypertension (chronic high blood pressure), a rate that remains stubbornly unchanged even as residents adopt healthier lifestyles post-COVID. What if technology supported users in their dietary management journeys?

We propose MealVitals, a cloud-native Software-as-a-Service application enabling users to track their nutrition intake. MealVitals generates meal plans and produces grocery lists, supporting users in every step of their dietary management journeys.

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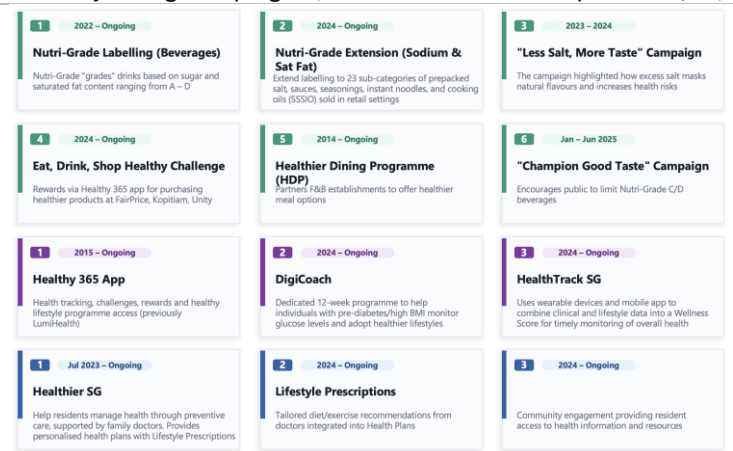
1. Business Case

1.1 Context & Problem Statement

The 2024 National Population Health Survey[1] conducted by the Ministry of Health (MOH) reports that about one in three Singapore residents continue to have hyperlipidaemia and hypertension, despite a rise in the percentage of residents engaged in sufficient total physical activity from 75% to 85% from 2022. Coupled with a rising obesity prevalence, these signs present an opportunity to help users with dietary management through market-to-table technology-enabled nutrition tracking.

To date, MOH has actively campaigned for healthy living through various campaigns:

Healthy Living Campaigns (Source: HPB Annual Report 2024/25)



MOH's initiatives to promote healthy living[2] primarily focus on:

1. Increasing awareness of healthy living through community events and online resources.
2. Helping residents make more informed choices through product labelling.
3. Providing generic diet recommendations (e.g. reducing sodium intake) through Healthy 365.

The evidence suggests that translating healthy living campaigns into better-informed dietary choices remains a challenge across the resident population. For instance, Chan et al. (2023) surveyed 955 Singaporeans and found substantial gaps

in residents' knowledge and ability in managing their salt and sodium consumption[3]:

1. Many were unaware of the recommended daily intake for salt (68%) and for sodium (83%) and struggle to interpret the sodium content of food products across different serving weights.
2. 30% of the participants reported that they were confused by the Nutrition Information Panel.

Consequently, residents often chose food based on "experience and knowledge" rather than by labels[3]. This suggests that residents are not making informed choices when planning their meals. Even when residents can correctly identify the healthier dietary choice, the cognitive effort required to apply this knowledge in the process leading to food consumption – meal planning, grocery shopping -- proves an obstacle towards adherence. There is a clear opportunity for a solution that simplifies dietary decision-making.

1.2 Proposed Solution

MealVitals addresses this gap by providing a cross-platform meal planning service that converts dietary constraints into personalised meal plans and grocery lists targeting the following markets:

1. The 1.1 million residents aged 18 and older diagnosed with or at risk of chronic conditions. These users typically have some nutritional knowledge but lack the resources to consistently make optimal dietary choices.
2. The caregivers planning meals for people with the conditions stated above.

MealVitals is designed to support users by generating personalised meal plans based on health profiles, dietary constraints, and personal preferences. This helps to improve dietary management adherence by simplifying meal-planning and grocery-buying. In addition, studies[4] have shown that elderly residents with chronic conditions often struggle with digital tools, and MealVitals addresses this gap by enabling a digitally savvy caregiver to plan meals on their behalf. While MealVitals focuses on personalised meal recommendations, it does not provide medical advice, which remains the responsibility of healthcare professionals.

2. Business Model

2.1 Overview

MealVitals uses a freemium model, giving all users access to basic features for meal management & health monitoring at no cost. User personalisation such as customising meal plans and receiving dynamic recommendations are reserved for premium users. The table below summarises the features for each tier:

Feature	***Free Tier***	***Premium Tier***
Profile Limit	Maximum of 5 profiles	
Health & Dietary Filters	✓	✓
Meal Actions (swap, remove, favourite)	✓	✓
Grocery Cart & Shopping List	✓	✓
Personalised Health Dashboard	✓	✓
Health Measurements Logging	✓	✓
Meal Selection	Pre-filtered based on existing health profile, dietary preferences and allergies	Personalised recommendations that adapt to the user's goals and eating history
Meal Plan Creation	Manual creation	Plan by day or week with automation

2.2 Competitive Analysis

Name	Monthly Fee	Dish Recognition	Nutrition Analysis	Meal Log	Meal Plan	Grocery List
Healthy 365 (MOH)	Free	Yes	Yes	Yes	No	No
Food (Ig) NUS-ISS	Free	Yes	Yes	Yes	No	No
Ventrickle	$13 /month	Yes	Yes	Yes	No	No
MealVitals	$9.9 /month	**No**	Yes	Yes	**Yes**	**Yes**

MealVitals offers two unique features. The first is intelligently planning meals based on constraints – dietary restrictions, nutritional targets, user preferences, and user recent meals – the second being in generating grocery lists. These advantages allow MealVitals to support users in their entire dietary management journey, from market to table.

MealVitals will not offer Dish Recognition as a feature initially, but subscribing to the FoodAI API[5] for food image recognition, which is the same API that Healthy 365 consumes, will allow MealVitals to quickly achieve feature parity with its competitors.

2.3 Cloud vs On-Premises Cost Comparison

We recommend using a cloud deployment for this project for three main reasons: zero upfront CAPEX investment, seamless scalability, and a lower total cost of ownership (TCO).

2.3.1 Target Audience and Usage Projections

Our target audience of 1.1 million user is detailed in Section 1.2. Assuming a 2% adoption rate over three years: the total addressable base is 22,000 users at full maturity. Uptake is projected to ramp from 20% in Year 1 (4,400 MAU) to 65% in Year 2 (14,300 MAU), reaching 100% by Year 3 (22,000 MAU).

We also assume an activity pattern concentrated within a 12-hour window, peaking during that period and gradually fading, while the remaining 12 hours are largely idle. This is because our users are based entirely in Singapore within a single time zone.

2.3.2 Cost Components and Assumptions

On-Premises Deployment: We consider both Capital Expenditure (CAPEX) and Operating Expenditure (OPEX). CAPEX includes the initial investment for servers, network routers, and other necessary data center hardware. For this estimate, CAPEX covers two server racks deployed in a location in Singapore and depreciates over three years. We must provision Year 1 to serve the maximum projected traffic of Year 3. OPEX includes ongoing electricity, internet subscriptions, and facility overheads.

Cloud Deployment: As noted by Lenovo Study[10], cloud deployment incurs only OPEX via ongoing subscription fees (e.g., compute, storage). CAPEX is not required as the model is purely pay-as-you-go.

Note: Staffing costs are excluded from this discussion, as the project team will adopt a "start-up mindset", contributing their time and effort as part of the core team

2.3.3 Three-Year Cost Comparison

The Total Cost of Ownership (TCO) for the cloud is 28% lower than an on-premises deployment over a three-year period, as seen in Table 3. The primary driver is the lack of upfront CAPEX for the cloud option. Details of the on-premises deployment cost structure can be found in Table 1 and Table 4, while Table 2 for that of Cloud option deployed on Google Cloud Platform (GCP).

Table 1: Monthly cost of on-premises deployment

Component	Monthly Cost ($)
CAPEX Amotisation	$272
Electricity	$96
Broadband subscription	$150
maintenance and others	$50
Total	**$568**

Table 2: Monthly cost of Cloud deployment (Year 2)

Component	Monthly Cost ($)
Compute - Backend	$25
Compute - Website	$25
Compute - Recommender function	$20
Database	$323
Total	**$393**

Table 3: 3 Year TCO

Option	3-Year TCO
On-premises	$20,458
Cloud	$15,180

***Table 4:* on-premises CAPEX upfront cost**

Component	CAPEX Cost ($)
Compute (server)	$3,800
Extra Storage	$2,000
Network router	$2,000
Power supply	$1,000
Rack switches	$1,000
Total	**$9,800**

Software cost here is free as we assume the adoption of open-source OS (ubuntu/Debian).1 server rack here assumes a low-end performance server Dell PowerEdge R730 Server with 1 extra hard disk, and a router and rack switches.

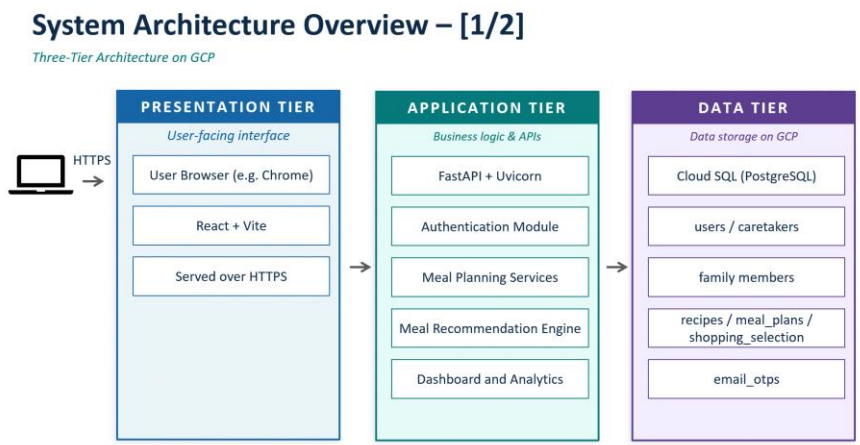
The monthly cloud cost scales annually alongside user growth. Conversely, an on-premises setup requires us to provision for Year 3 capacity starting in Year 1, as hardware upgrades are difficult to implement mid-cycle. Notably, while the cloud's monthly cost eventually exceeds the on-premises monthly cost in Year 3 due to high traffic volume as observed in Figure 1, the cumulative three-year savings remain significant.

Fig 1: Average monthly cost comparison - Cloud option adjusts the provision as users

3. Architecture & Implementation

3.1 System Architecture & Data Modelling

Application Design



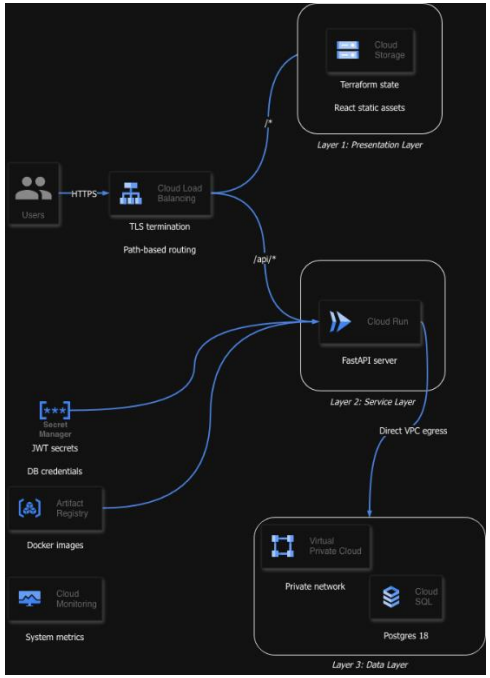
The application follows a three-tier architecture

Presentation layer – React SPA application built with Vite and delivered over HTTPS

Application layer - Fast API application exposes REST APIs, handles authentication and business logic, running on Uvicorn

Database layer: PostgreSQL using SQL Alchemy for ORM and Alembic for schema migrations

System Design (The application is deployed on GCP)



Cloud DNS – Resolves hostname "mealvitals.app" to Cloud Load Balancer public IPv4 address

Cloud Load Balancer – Implements TLS termination for simpler downstream request handling; Performs path-based routing, directing index requests to the static site hosted on Cloud Storage and API requests to Cloud Run

Cloud Storage – Hosts static website; Acts as bucket for the application backend; Stores IaC state files

Cloud Run – Serverless compute handling API services and Auth services; Retrieves secrets from Secret Manager and builds from images on Artifact Registry

Cloud SQL – Stores application data and auth data; Deployed within VPC to avoid exposing public IP address

Data Modelling

The data model comprises six entities

1. Individuals: user accounts
2. Households: representing family members of the user
3. Meals: recipes, ingredients, nutritional information (crawled from MOH)
4. Meal Plans: schedules of meals
5. Shopping Selections: aggregated ingredient data & nutritional values
6. User Authentication: tokens for email verification

3.2 Meal Recommendation Algorithm

The meal recommender is built on Google OR-Tools CP-SAT[7], which serves as the optimization engine of the planning system. In MealVitals, premium users receive personalised meal recommendations that account for dietary preferences, allergies, and health conditions. The recommender generates meals for breakfast, lunch, and dinner across planning horizons of 1 to 7 days. Users can preserve accepted meals as fixed assignments and generate recommendations only for selected slots, making the system suitable for interactive planning. This is a constraint optimization[6] problem, and CP-SAT is well-suited to it because it handles binary and integer decisions under both hard constraints and soft objective trade-offs.

The optimization algorithm is defined by the meal slots to be filled, the candidate recipes available, and the user-specific constraints that govern selection.

Component	Description
Demand	Meal slots to be filled across 1 to 7 days for breakfast, lunch, and dinner
Supply	Candidate recipes with per-serving nutrient values, eligible meal types, and health-suitability labels
User constraints	Nutritional targets, dietary preferences, allergies, medical conditions, and thresholds defined at runtime
Pre-processing	Recipes that violate allergy, diet, or medical suitability rules are filtered out before optimization

For each day *d*, meal slot *m*, candidate recipe *r* and nutrient type *k*, the solver determines whether the recipe is selected and how many servings are assigned to it. The daily nutrient total is computed as the sum of each selected recipe's per-serving nutrient value multiplied by its assigned servings:

$$\text{DailyNutrient}_{d,k} = \sum_m \sum_r \text{Nutrient}_{r,k} \times \text{Servings}_{d,m,r}$$

This allows the system to optimize the nutritional effect of the meal plan as a whole rather than score recipes independently. The main objective is to minimize deviation from the user's target intake, especially for calories, protein, and carbohydrates. Additional penalties discourage exceeding fat, sodium, and sugar targets, repeating recipes too often, selecting cautionary dishes, and reusing recently served meals.

The algorithm can also take nutrient thresholds as runtime inputs during generation. When provided, these override the default profile-derived targets and are enforced as hard constraints:

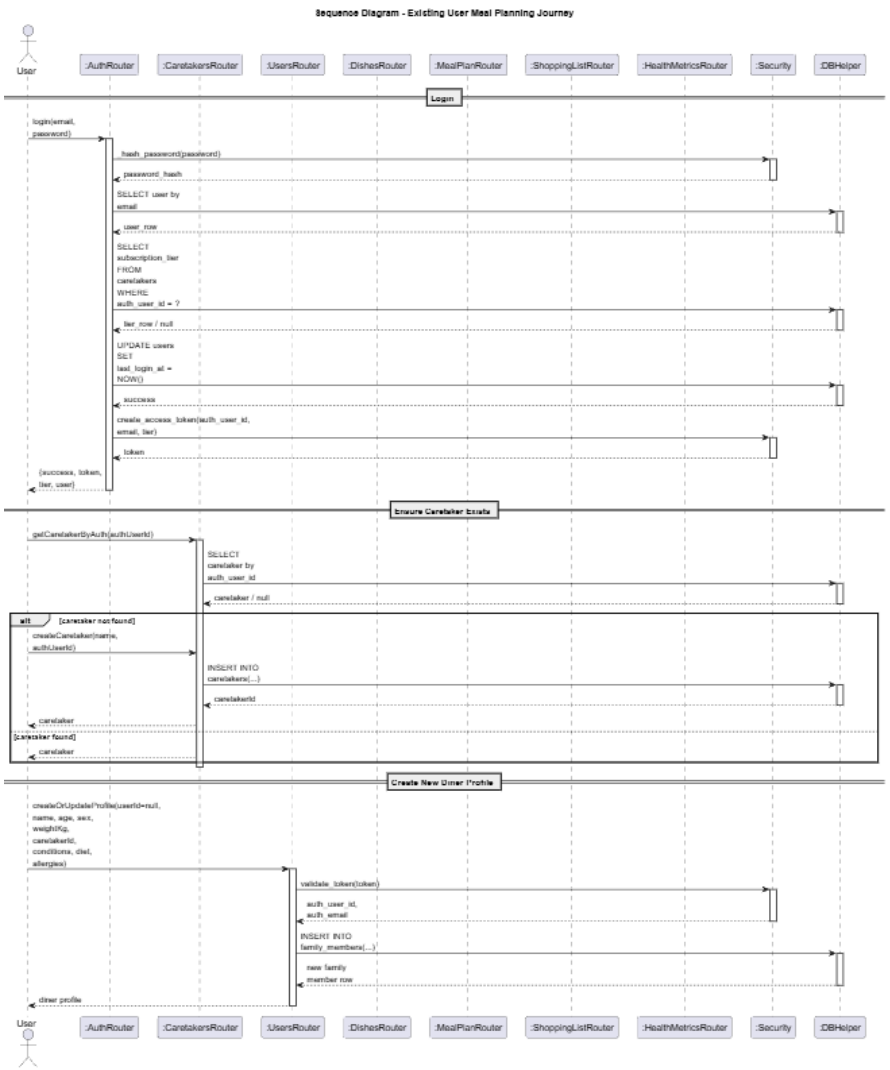
$$\text{DailyNutrient}_{d,k} \leq \text{Threshold}_k$$

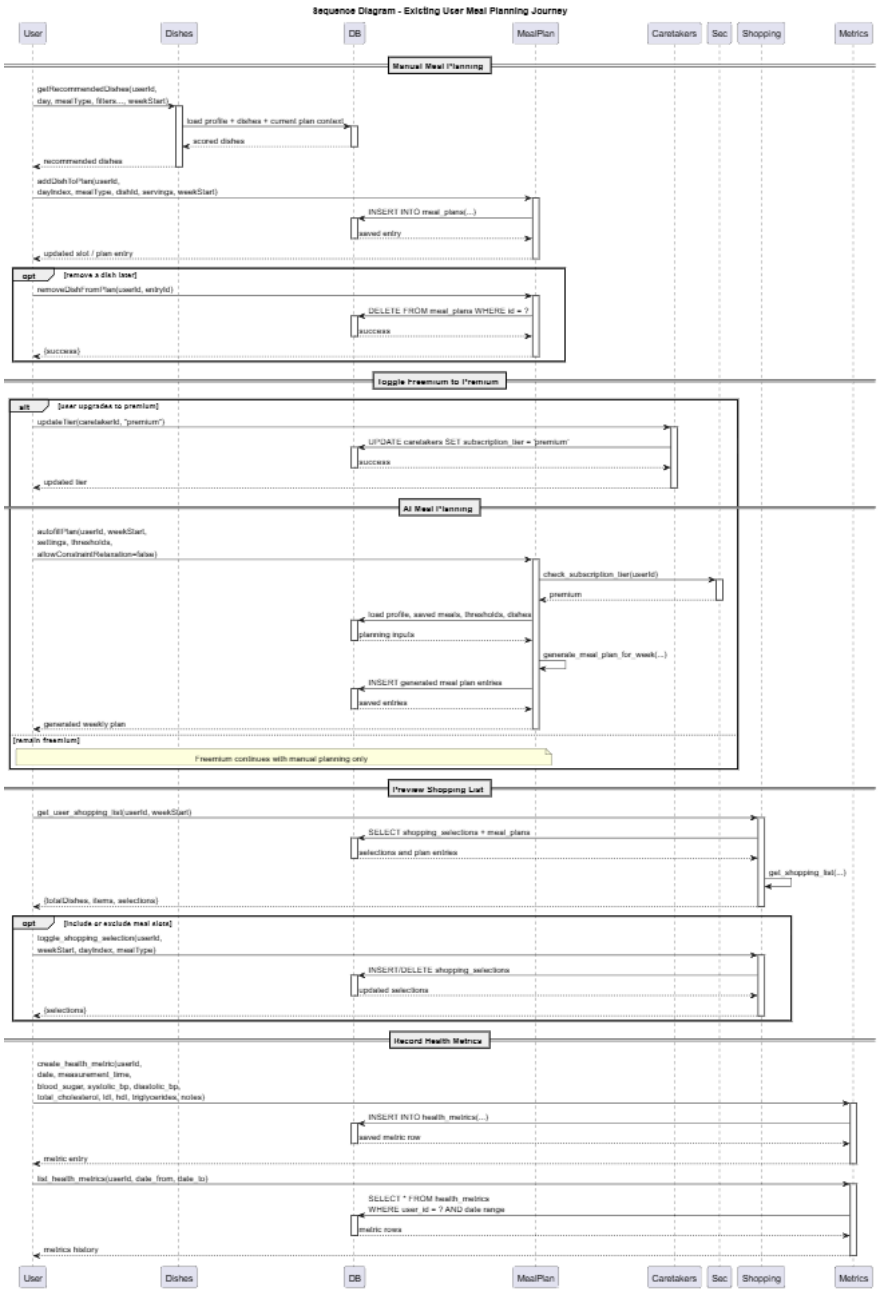
The algorithm also includes structural feasibility constraints to ensure the output is usable in the application. Each meal slot must contain a valid number of dishes, servings must remain within allowed bounds, and existing saved meals may be fixed during autofill so that only the remaining gaps are filled.

Overall, CP-SAT is a good fit because it supports global optimization over the full meal plan. A simpler scoring-based method would struggle to handle multiple interacting requirements, such as nutritional targets, dietary restrictions, preserved meals, partial-week generation, dish-count rules, and variety constraints, within a single consistent framework.

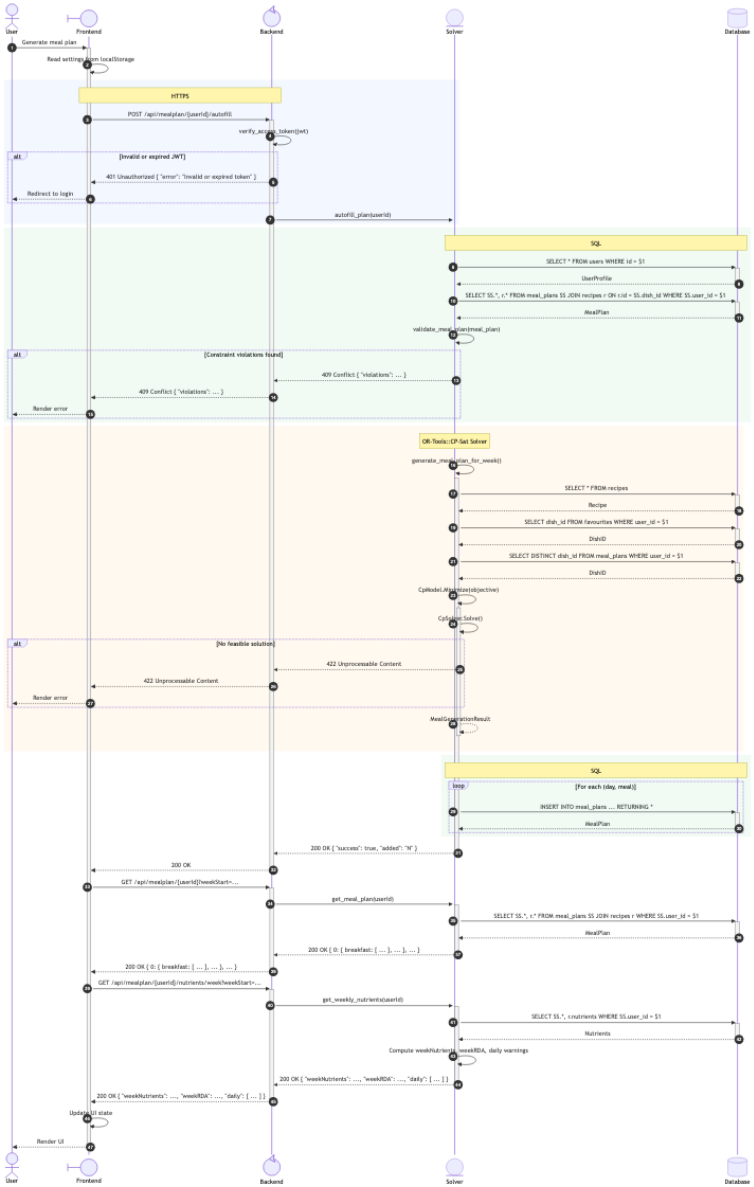
3.3 System Interaction and Behavioural Design

MealVitals supports 4 distinct use cases from 1) managing health profiles, 2) generating meal plans, 3) retrieving shopping list and 4) tracking health metrics – total nutrition value consumed and health monitoring.





A deep-dive into the CP-SAT solver is plotted below:



3.4 Security, Authentication and Secret Management

MealVitals handles sensitive user data such as email addresses, dietary preferences, allergies, and health-related profiles. As a multi-user SaaS platform, it is critical to ensure that only authenticated users can access the system and that each user's data is securely isolated for system integrity. The security model follows a five stage process:

Stage	***Purpose***	***Implementation Details***	***Why this matters?***
1. Email Verification	Ensures that all accounts correspond to valid and reachable users, reducing mass registration and improving overall system integrity	One-time verification code that are short-lived, single-use, and rate-limited Account remains inactivated until verified	Avoid mass registration using invalid or random email addresses Build a trusted communication channel for future operations such as password resets and notifications
2. JWT Issuance	Adopts JSON Web Token (JWT) based authentication to support its browser-based single-page application architecture Upon successful login, the backend issues a signed JWT containing user identity (sub) and expiry (exp) claims.	JWT signing keys are stored securely using GitHub Secrets and injected during the CI/CD pipeline, never hardcoded in the application Signing secret consists of a header, payload, and signature, where the signature is generated using a server-side secret.	Protecting the JWT signing secret is critical, as exposure of this key would allow attackers to forge valid authentication tokens Prevents accidental exposure through source code repositories and supports better separation between code and configuration
3. Request Handling	*All privileged endpoints enforce authentication before executing business logic*	*Bearer token required on all protected endpoints before it is processed* *Nginx enforces CORS by restricting allowed origins to the official frontend domain*	*Ensures that users must be in a valid session before accessing resources such as meal plans, shopping lists, or household data.* *Prevents unauthorised browser-based applications from making API requests on behalf of users*
4. Token Validation	The backend validates the token on every request, ensuring that it is properly signed, not expired, and associated with a legitimate session	Signature and expiry checked per request by backend middleware Invalid or expired tokens return	Any modification to the token payload invalidates the signature, ensuring that tokens cannot be tampered with or forged by clients
5. Authorisation	Following authentication, authorisation checks are applied to confirm that users can only access their own data or explicitly linked household members	Backend restricts resource access to the user's own data and explicitly linked household members. Server-side enforcement applied, as frontend restrictions alone are insufficient and can be bypassed.	This enforces strict per-user data isolation within the shared SaaS infrastructure.

4. Evaluation

4.1 User Engagement & Evaluation

The growth of MealVitals can be evaluated across three main funnels: acquisition, retention and monetisation[8]. Acquisition uses leading indicators to track whether new users discover and onboard onto MealVitals. Retention uses engagement and guardrail metrics to monitor user usage and behaviour over time. Lastly, monetisation uses lagging indicators to assess the health value delivered to users. The table below summarises the metrics to be collected through Google Analytics.

North Star: Number of users generating meal plans per week		
Leading Indicators (3): ***Signal proportion of target users that have yet to onboard with MealVitals***	Engagement (3) & Guardrails (2): ***Monitor MealVitals's usage and ensure users remain satisfied***	Lagging Indicators (1): ***Evaluate the health value that is delivered to users***
1. Health Profile Onboarding # new users completing health profile within 24 hours of sign up per week 2. First Meal Plan Recommended # new users generating meal plans within 48 hours of sign up per week 3. Meal Plan Acceptance & Adjustment # of meal plans accepted and/or adjusted per week	E1. 30/60/90 Days User Retention # users remaining active* for 30/60/90 days after onboarding E/G2. Thumbs-up, Thumbs-down Rating System Does the user like/dislike daily recommendations G3. Meal Plan Rejection # of meal plans rejected due to incompatibility with medically-advised diet	1. Improved Cardiometabolic Health e.g. # active users with reduced cholesterol / reduced blood pressure and/or improved glycaemic control after 6 months Note: Active users receive a twice-yearly survey assessing cardiometabolic improvement and app usability through automated via **Google Cloud Scheduler**

*users who have generated at least one meal plan in the past week

Active monitoring of these metrics signals the type of product work[9] needed to grow MealVitals and allows timely interventions such as user surveys or focus group discussions when any metric falls below the desired target.

4.2 System Performance Monitoring Plan

The user metrics above are meaningful only when the system is operating reliably. Rather than configuring alerts and dashboard manually via GCP Console, Terraform allows the configuration alerts and dashboard setup to be declared as code. This ensures that alert threshold and dashboards are reproducible and not subject to manual drift. The monitoring plan covers the following:

- Availability: website and API uptime
- Performance: meal plan generation latency
- Reliability: API error rate, CP-SAT solver failure logging

Resource Type	***Purpose***
google_monitoring_uptime_check_config	Perform an uptime check on MealVitals every 60 seconds
google_monitoring_alert_policy (error, latency)	Monitoring alert when - 5% of API requests are failing over a period of 60 seconds. - Meal plan generation took more than 5 secs to plan for 1 day over a period of 60 seconds
google_logging_metric	To log meal planning solver failure whenever API return empty meal plan
google_monitoring_notification_channel	Alert is sent to operation team via email when a policy violation is detected.
google_monitoring_dashboard	Dashboard to monitor availability, performance and reliability of the app

Terraform workflow applied to Monitoring

Step 1 — Write: *$ terraform init*

What it does: Running terraform init downloads the required provider plugins from the Terraform Registry.

Step 2 — Plan: $ terraform plan

What it does: Terraform reads the configuration, compares it against the current state file, and computes the exact changes it would make — which resources would be created, modified, or destroyed. It displays this plan to the engineer. It also validates syntax errors. This is non-destructive — terraform plan never makes any changes to real infrastructure. It is a read-only preview.

Step 3 — Apply: $ terraform apply

What it does: After confirmation, Terraform provisions the resources and updates the state file to record the new current state.

5. Limitations and Recommendations for Future Works

Data Coverage The meal dataset is not contextualised to Singapore's food culture. Future work could consider including local dishes by enabling community sharing, e.g. allowing users to submit recipes for their favourite local dishes or share their favourite dishes to included on the app, with AI used to generate nutritional value and recipes from the images/ text.

Secondary Market MealVitals address meal planning for individuals who home cooks, however, there are significant portion of Singaporeans who prefer eating out. Future work could extend the app's relevancy to include nearest hawker centre locator to better reflect Singapore's eating out culture.

New Features Future development could explore integrating with local supermarket e.g. NTUC Fairprice for direct cart-out functionality. This allows users to purchase ingredients remotely, streamlining the journey from meal planning to grocery checkout.

Technical MealVitals's meal recommendation generate meal recommendations sequentially across multiple days. Further work could consider to parallelise this process to improve performance and better balance computational load as user usage increases

Security The current approach to secret management is adequate and significantly more secure than hardcoding credentials. Further improvements could further enhance this by including runtime secret retrieval from a dedicated cloud secret manager.

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